CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V[th] **Semester of B Sc Biochemistry   Examination March 2019**

Course code & Name: BC503 Metabolism-I

Date: 19/03/2019   Day: Tuesday   Time: 10:00 a.m To 12:00 noon   Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. **10**

1. α -oxidation of fatty acids occurs in ________.

(i) mitochondria
(ii) peroxisomes
(iii) endoplasmic reticulum
(iv) cytoplasm

2. Ketone bodies are over produced during ________________.

(i) Diabetes
(ii) Starvation
(iii) Both of the above
(iv) None of the above

3. Arachidonic acid leads to formation of _______________.

(i) prostaglandins
(ii) leukotrienes
(iii) thromboxanes
(iv) all of the above

4. Mammals cannot utilize fatty acids as sole source of energy due to lack of ________.

(i) Pyruvate dehydrogenase complex (ii) lactic acid fermentation

(iii) Electron Transport Chain (iv) Glyoxylate cycle

5. NAD-linked dehydrogenases oxidize the substrate be removing ____________.

(i) hydride ion (ii) direct electron

(iii) hydrogen atom (iv) All of the above

6. Chief role of non-oxidative pentose phosphate pathway is _______________

(i) NADPH synthesis (ii) Biosynthesis of ribose-5-P

(iii) recycling ribose-5-P to produce glucose-6-phosphate (iv) recycling glutathione

7. Which of the following is NOT TRUE for glycogen synthase?

(i) It requires UDP-glucose as substrate (ii) It cannot make the ($\alpha 1 \rightarrow 6$) bonds

(iii) It cannot initiate a new glycogen chain *de novo* (iv) It is also known as UDP-glucose pyrophosphorylase

8. Which of the following is the essential intermediate in palmitate biosynthesis?

(i) HMG-CoA (ii) Mevalonate

(iii) Acetyl-CoA (iv) Malonyl-CoA

9. Lactic produced in muscle cells during extensive exercise is metabolized in liver by _____.

(i) Cori's cycle (ii) anaplerosis

(iii) cellular respiration (iv) oxidative pentose phosphate pathway

10. Which of the following is/are TCA cycle enzyme(s)?

(i) α-ketoglutarate dehydrogenase (ii) isocitrate dehydrogenase

(iii) Fumarase (iv) All of the above

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Instructions:
1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I

Q - II Answer the following questions as directed **10**

(A) Outline the mechanism of digestion, mobilization and transport of fats. **04**

OR

(A) Discuss the oxidative and non-oxidative phases of pentose phosphate pathway. **04**

(B) **Write a short note on the following (Any TWO)** **06**

(i) Importance of phosphorylated intermediates in glycolysis

(ii) Fatty acyl Co-A synthetase

(iii) Pyruvate carboxylase and Pyruvate dehydrogenase complex

SECTION - II

Q-III Answer the following questions as directed **15**

(A) **Explain the following (Any FIVE)** **15**

(i) Preparatory phase of glycolysis
(ii) Enzymes of anaplerosis
(iii) Feeder pathways of glycolysis
(iv) Overview of glycogen catalysis
(v) Major steps of cholesterol biosynthesis
(vi) Cori's cycle
(vii) Ketone bodies and their biosynthesis
(viii) β-oxidation of fatty acids